GRANT NEILSON

Bioinformatician with six years of industry experience analyzing multiple NGS data types, with a specialism in whole genome sequencing (WGS). Highly proficient in R and Python, with a strong track record of developing analytical tools and packages for high-dimensional omics data. Passionate about building robust analytical workflows and contributing to open-source repositories to drive forward scientific research. An articulate and reliable individual with excellent communication skills and a strong track record of effective collaboration across multidisciplinary teams.

INDUSTRY EXPERIENCE

2026 | 2023

Senior Computational Biologist

CoSyne Therapeutics

London, UK

- Architected and deployed containerized bioinformatics infrastructure using Docker and AWS ECR, establishing scalable cloud-native solutions for multi-omic data analysis. Designed and implemented CI/CD pipelines for reproducible computational workflows, reducing deployment time by 60%.
- Lead developer and maintainer of multiple internal Python and R packages, establishing comprehensive testing frameworks, documentation standards, and code review processes. Implemented project templating system using Cruft, standardizing development practices across the organization.
- Engineered hyperscale cloud infrastructure on AWS for processing ~3 million single-cell sequencing samples, implementing automated alignment and QC pipelines. Optimized compute resource allocation and parallelization strategies, achieving 50% cost reduction and 30% performance improvement.
- Contributed to open-source NF-core Sarek pipeline (v3.4.1), implementing performance optimizations and workflow improvements. Developed custom Nextflow modules for variant calling and quality control processes.
- Established software engineering best practices including version control workflows, code review standards, unit testing frameworks, and comprehensive documentation. Mentored team members on software development methodologies and DevOps practices.

2023 | 2020

Computational Biologist

Fios Genomics

Edinburgh, UK

- Directed projects focusing on high-dimensional omics data types (WES, WGS, scRNA-seq, proteomics), delivering comprehensive analyses and reports for clients.
- Developed in-depth knowledge of statistical methodologies, mastering model selection and fitting techniques, including SVM, elastic net, random forest, and linear/logistic regression, while accounting for covariates.
- Conducted complex regression model analyses for biomarker discovery in bladder and lung cancer.
- Actively contributed to internal R&D by developing and maintaining specialized R packages, enhancing the team's analytical capabilities.
- Led client calls, effectively communicating complex analytical results, ensuring clarity and client understanding.

View this CV online with links at https://grantn5.github.io/cv/cv_engineer.html

CONTACT

- grant.neilson5@gmail.com
- GitHub
- LinkedIn

LANGUAGES

- R
- Python
- Bash
- Nextflow

TECHNOLOGIES

- GitLab/GitHub
- Docker
- aws
- Azure

OPEN SOURCE & PUBLIC PROJECTS

- Sarek
- bixverse
- mainfoldsR

Made with the R package pagedown.

The source code is available on GitHub.

Last updated on 2026-02-17.

2020 | 2019

Computational Biologist

Complex Diseases Epigenetics group, Exeter Medical School — Exeter, UK

- Developed novel bioinformatics pipelines for Epigenome/Genome wide association studies, mQTL analysis and quality control of Illumina EPIC array data.
- Took the lead on a project performing EWAS and GWAS analysis in ADHD patients. This involved fitting linear and logistic regression models while accounting for covariates.
- Was selected for a secondment to New York to work at the Mount Sinai medical school as a visiting bioinformatician. This was unfortunately cancelled due to COVID-19.

EDUCATION

2019 | 2015

MSCi Biology

University of Southampton — Southampton, UK

- **Grade** : First Class Honors
- Winner of the British Society for Immunology Graduate of the year award.
- Recipient of the summer studentship from the Genetics Society (2017).

2015 | 2013

School

Merchiston Castle Shool — Edinburgh, UK

- A Levels: Biology, Chemistry, Maths

SELECTED PUBLICATIONS

2025

Multi modal single cell foundation models via dynamic token adaptation

BioRxiv

- Wenmin Zhao, AnaSolaguren-Beascoa, **Grant Neilson**, Regina Reynolds, Louwai Muhammed, Liisi Laaniste, Sera Aylin Cakiroglu
- Role: Data Preproccessing
- doi: https://doi.org/10.1101/2025.04.17.649387

2025

Detecting cell level transcriptomic changes of Perturb seq using Contrastive Fine tuning of Single Cell Foundation Models

BioRxiv

- Wenmin Zhao, AnaSolaguren-Beascoa, **Grant Neilson**, Regina Reynolds, Louwai Muhammed, Liisi Laaniste, Sera Aylin Cakiroglu
- Role: Data Preproccessing
- doi: https://doi.org/10.1101/2025.04.17.649395